

THE UNIVERSITY OF BIRMINGHAM
DEPARTMENT OF CIVIL ENGINEERING

Mechanics 1

Tutorial Sheet - 1

1.

$$H = 15 + 10 \cos 60^\circ$$

$$= 20 \text{ kN}$$

$$V = 10 \sin 60^\circ$$

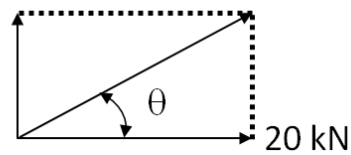
$$= 8.66 \text{ kN}$$

$$R = \sqrt{H^2 + V^2}$$

$$= 21.8 \text{ kN}$$

$$\theta = \tan^{-1} \left(\frac{V}{H} \right) = 23.4^\circ$$

$$8.66 \text{ kN}$$



2. $R = 10 + 8 \cos 60^\circ - 12 \cos 30^\circ - 20 \cos 55^\circ$

$$= -7.864 \text{ kN}$$

$$V = +8 \sin 60^\circ + 12 \sin 30^\circ - 20 \sin 55^\circ$$

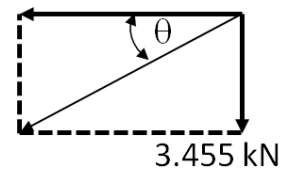
$$= -3.455 \text{ kN}$$

$$R = \sqrt{H^2 + V^2}$$

$$= 8.6 \text{ kN}$$

$$\theta = \tan^{-1} \left(\frac{V}{H} \right) = 23.7^\circ$$

$$7.864 \text{ kN}$$



3.

(a)

$$\sum R = 0$$

$$R_{A,H} + 7x \cos 60^\circ + 8x \cos 45^\circ = 0$$

$$R_{A,H} = -9.2 \text{ kN}$$

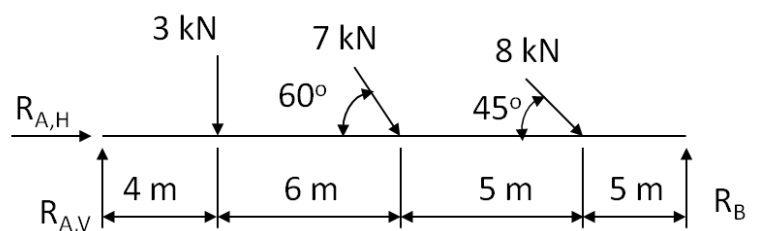
$$\begin{aligned} +\sum M_A &= 0 \\ R_B \times 20 - 3 \times 4 - 7 \sin(60^\circ) \times 10 \\ &\quad - 8 \sin(45^\circ) \times 15 = 0 \end{aligned}$$

$$R_B = 7.9 \text{ kN}$$

$$\sum V = 0$$

$$R_{A,V} + 7.9 - 3 - 7 \sin(60^\circ) - 8 \sin(45^\circ) = 0$$

$$R_{A,V} = 6.9 \text{ kN}$$



(b)

$$\sum R = 0$$

$$R_{A,H} = 0 \text{ kN}$$

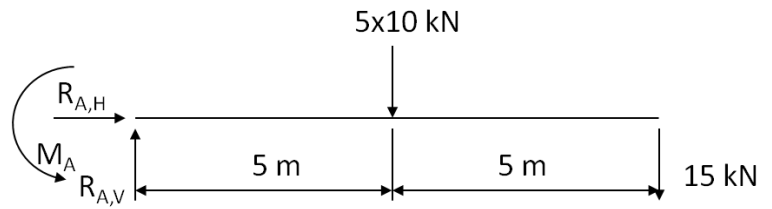
$$+\sum M_A = 0$$

$$M_A - 50 \times 5 - 15 \times 10 = 0$$

$$\sum V = 0$$

$$R_{A,V} - 15 - 50 = 0$$

$$R_{A,V} = 65 \text{ kN}$$



(c)

$$\sum R = 0$$

$$R_{A,H} - 20 = 0$$

$$R_{A,H} = 20 \text{ kN}$$

$$+\sum M_B = 0$$

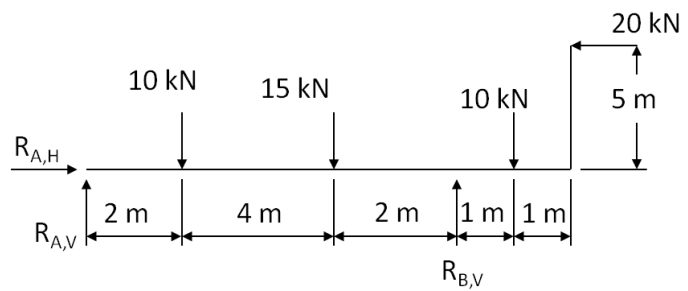
$$R_{A,V} \times 8 - 10 \times 6 - 15 \times 2 + 10 \times 1 - 20 \times 5 = 0$$

$$R_{A,V} = 22.5 \text{ kN}$$

$$\sum V = 0$$

$$22.5 + R_{B,V} - 10 - 15 - 10 = 0$$

$$R_{B,V} = 12.5 \text{ kN}$$



(d)

$$\sum R = 0$$

$$R_{A,H} = 0 \text{ kN}$$

$$+\sum M_A = 0$$

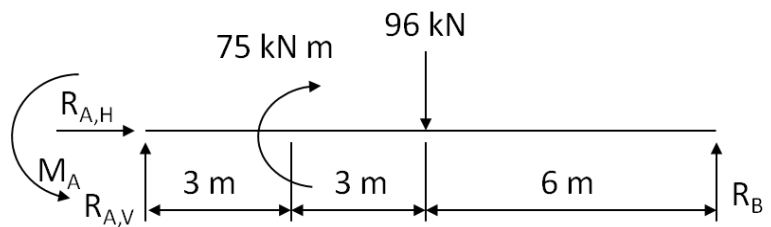
$$12R_B - 96 \times 6 - 75 = 0$$

$$R_B = 54.2 \text{ kN}$$

$$\sum V = 0$$

$$R_{A,V} + 54.2 - 96 = 0$$

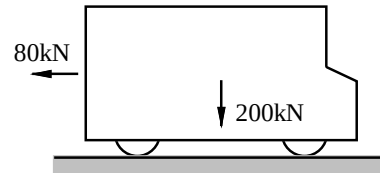
$$R_{A,V} = 41.8 \text{ kN}$$



4. Resultant, $R = (200^2 + 80^2)^{0.5} = 215.4 \text{ kN}$

$\alpha = \tan^{-1}(200/80) = 68.2^\circ \text{ to the horizontal, or}$

$21.8^\circ \text{ to the vertical}$



5. Resolve horizontally:

$$T_{AB} \cos \alpha - T_{BC} \cos \beta = 0$$

$$T_{AB} = T_{BC} \frac{\cos \beta}{\cos \alpha} \quad (1)$$

Resolve Vertically:

$$T_{AB} \cos(90 - \alpha) + T_{BC} \cos(90 - \beta) - 50 = 0$$

$$T_{AB} \sin \alpha + T_{BC} \sin \beta = 50 \quad (2)$$

Combining (1) and (2)

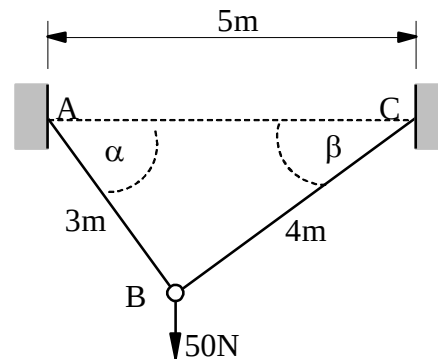
$$T_{BC} \left\{ \cos \beta \frac{\sin \alpha}{\cos \alpha} + \sin \beta \right\} = 50$$

$$T_{BC} \{ \cos \beta \tan \alpha + \sin \beta \} = 50$$

$$T_{BC} \left\{ \frac{4}{3} \frac{4}{5} + \frac{3}{5} \right\} = 50$$

$$\frac{25}{3} T_{BC} = 250 \quad \Rightarrow \quad T_{BC} = 30 \text{ N}$$

$$T_{AB} = 40 \text{ N}$$

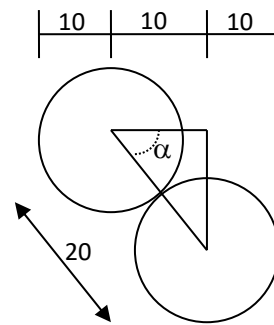


$$\sin \alpha = \frac{4}{5} \quad \cos \alpha = \frac{3}{5} \quad \tan \alpha = \frac{4}{3}$$

$$\sin \beta = \frac{3}{5} \quad \cos \beta = \frac{4}{5}$$

NB since $\angle ABC = 90^\circ$, resolving in the direction of AB would give the answer for T_{AB} direct, ditto for the direction of BC and T_{BC} .

6i) $\cos\alpha = 10/20$. $\alpha = 60^\circ$.

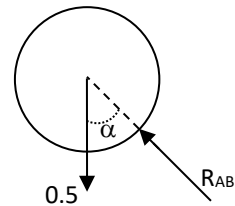


6ii) Consider the equilibrium of the upper coin.

Resolve vertically:

$$0.5 - R_{AB}\sin\alpha = 0$$

$$\therefore R_{AB} = \frac{1}{2} \frac{1}{\sin\alpha} = \frac{1}{2} \frac{2}{\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ N}$$



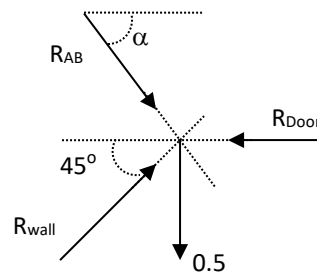
6iii) Consider the equilibrium of the lower coin.

Resolve vertically:

$$R_{\text{wall}}\cos 45^\circ - R_{AB}\sin\alpha - 0.5 = 0$$

$$\frac{1}{\sqrt{2}} R_{\text{wall}} = 0.5 + R_{AB} \sin\alpha = 0.5 + \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2}$$

$$R_{\text{wall}} = \sqrt{2} \text{ N}$$



Resolve horizontally:

$$R_{AB}\cos\alpha + R_{\text{wall}}\cos 45^\circ - R_{\text{Door}} = 0$$

$$R_{\text{Door}} = \frac{1}{\sqrt{3}} \frac{1}{2} + 1 = 1 + \frac{1}{2\sqrt{3}} \text{ N}$$

- 7 Take moments about A (+ve anticlockwise)

$$8V_E - 2 \times 10 - 6 \times 25 \sin 60 - 3 \times 25 \cos 60 = 0.$$

$$8V_E = 20 + 150 \frac{\sqrt{3}}{2} + 37.5 \quad V_E = 23.43 \text{ kN}$$

Take moments about E (+ve clockwise)

$$8V_A - 6 \times 10 - 2 \times 25 \sin 60 + 3 \times 25 \cos 60 = 0.$$

$$8V_A = 60 + 43.3 - 37.5 \quad V_A = 8.22 \text{ kN}$$

Check: Resolve vertically: $V_A + V_E = 10 + 25 \sin 60 = 31.65 \text{ kN}$

$$V_A + V_E = 23.43 + 8.22 = 31.6, \text{ hence okay}$$

Resolve horizontally: $25 \cos 60 - H_A = 0$, $H_A = 12.5 \text{ kN}$

- 8 $V_A + V_B + V_C = 360$

Take moments about line BC: $9V_A - 3(100 + 260) = 0$, $V_A = 120 \text{ kN}$

Take moments about a line through B and perpendicular to BC:

$$(100 \times 3) - (3 \times V_A) + (260 \times 4.5) - (V_C \times 6) = 0$$

$$V_C = 185 \text{ kN}$$

$$V_B = 55 \text{ kN}$$

9 Resolve vertically $V_A + V_B = 50 + 30 = 80$

Take moments about A (+ve anticlockwise)

$$-2 \times 50 - 5 \times 30 + 7V_B + 15 - 65 = 0. \quad \Rightarrow \quad V_B = 42.86 \text{ kN}$$

Take moments about B (-ve anticlockwise)

$$-2 \times 30 - 5 \times 50 + 7V_A - 15 + 65 = 0 \quad \Rightarrow \quad V_A = 37.14 \text{ kN}$$

Check: $V_A + V_B = 37.14 + 42.86 = 80$, okay.
